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Original Article

Final Overall Survival and Molecular Data Associated with Clinical Outcomes in Patients Receiving Ipatasertib and Abiraterone in the Phase 3 IPATential150 Trial

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Abstract

Background and objective: In the phase 3 IPATential150 trial, ipatasertib addition to abiraterone significantly reduced the risk of disease progression in men with metastatic castration-resistant prostate cancer (mCRPC) with PTEN loss on immunohistochemistry (IHC), but not in the intention-to-treat (ITT) population. Here we report the final overall survival (OS) analysis and present results for prespecified and exploratory biomarker analyses.

Methods: Patients were randomized to receive ipatasertib (400 mg once daily) or placebo. All patients received abiraterone (1000 mg once daily) and prednisone (5 mg twice daily). OS was assessed in patients with PTEN loss on IHC and the ITT population. Exploratory biomarker analyses included *PTEN* status via next-generation sequencing (NGS) and other key genomic alterations.

Key findings and limitations: At final analysis (median follow-up 33.9 mo), ipatasertib addition did not improve OS for patients with PTEN loss in IHC (n = 521; stratified hazard ratio [sHR] 0.94, 95% confidence interval [CI] 0.76–1.17; p = 0.57) or the ITT population (n = 1101; sHR 0.91, 95% CI 0.79–1.07; not formally tested). Exploratory NGS assessments identified subgroups with genomic *PTEN* loss (n = 208) or *PIK3CA*/*AKT1*/*PTEN* alterations (n = 250), with potentially better outcomes from ipatasertib (HR 0.76, 95% CI 0.54–1.07; and HR 0.70, 95% CI 0.51–0.96, respectively). Limitations include the

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exploratory nature of the analysis, incomplete availability of NGS data, and potential intrapatient heterogeneity.

Conclusions and clinical implications: Ipatasertib addition to abiraterone did not improve OS for men with mCRPC, regardless of PTEN status on IHC. Exploratory biomarker analyses identified additional genomic alterations of potential clinical relevance for AKT blockade in mCRPC that require further validation in prospective studies.

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ADVANCING PRACTICE

What does this study add?

Addition of the AKT inhibitor ipatasertib to abiraterone and prednisone did not significantly improve overall survival for patients with previously untreated metastatic castration-resistant prostate cancer (mCRPC) with or without tumor PTEN loss as determined by immunohistochemistry. Exploratory next-generation sequencing assessments identified subgroups of patients with genomic *PTEN* loss and alterations in the PIK3CA/AKT1/PTEN pathway that could benefit from AKT inhibition. Differences in patient populations identified as having PTEN loss using different assays were observed, including varying *PTEN* mRNA expression levels and transcriptional profiles. In addition, exploratory biomarker analyses identified other genomic alterations beyond the PIK3CA/AKT1/PTEN pathway potentially associated with outcomes with ipatasertib. These exploratory analyses offer important insights into a better understanding of the regulation of AKT signaling and disease biology in mCRPC and may offer guidance for biomarker selection in future clinical trials assessing agents targeting this pathway.

Clinical Relevance

Advancing targeted therapies for prostate cancer necessitates comprehensive biomarker evaluations, ideally conducted in rigorously designed prospective studies. The IPATential150 trial previously demonstrated that patients with mCRPC who had tumors with PTEN loss by immunohistochemistry experienced improved radiographic progression-free survival with the addition of the AKT inhibitor ipatasertib to abiraterone compared with placebo plus abiraterone. However, in the final overall survival analysis, the addition of ipatasertib to abiraterone did not result in a survival benefit. Interestingly, exploratory analyses suggest that *PTEN* loss identified via next-generation sequencing, along with other genomic alterations in related pathways, may correlate with responses to AKT inhibitors in mCRPC. While these findings are hypothesis-generating, they lay the groundwork for future targeted trials aimed at advancing precision medicine for patients with mCRPC. Associate Editor: Gianluca Giannarini, MD.

Patient Summary

In this report we looked at whether adding ipatasertib to abiraterone and prednisone affected survival for men with metastatic prostate cancer that no longer responds to hormone therapy castration. The addition of ipatasertib did not improve survival. However, mutation analysis identified potential biomarkers that can inform the design of future clinical studies using this class of drug for the treatment of prostate cancer.

1. Introduction

Men with metastatic castration-resistant prostate cancer (mCRPC) have a median life expectancy of <3 yr [1–3]. Despite the availability of many classes of treatment for mCRPC, patients almost inevitably develop life-threatening, treatment-resistant disease [4]. mCRPC is a genomically heterogeneous disease, explaining much of the development of resistance to cancer therapy and difficulties in treating it [5–7]. This highlights the need for targeted therapies and biomarkers to guide treatment choices in mCRPC [8].

The AKT signaling cascade is upregulated in up to 50% of prostate cancers, which can occur because of a range of genomic alterations, most commonly loss of function of the tumor suppressor PTEN, which negatively regulates the AKT pathway [9]. Loss of PTEN function is observed in 40–50% of patients with mCRPC and is associated with poorer clinical outcomes [10–12]. Multiple factors can lead to loss of PTEN function, including genomic deletions and rearrangements, and nongenomic aberrations such as microRNA-mediated dysregulation and promoter methylation [10,13–15]. Detection of gene deletion via fluorescence in situ hybridization and evaluation of protein loss via immunohistochemistry (IHC) have been the predominant ways for detecting PTEN loss, each with their own benefits and limitations [10,16]. Despite technical differences, the two assays have demonstrated good concordance of up to 80%, although defining the optimal cutoff for PTEN loss

remains challenging [13,17–19]. In addition, next-generation sequencing (NGS) assessment of *PTEN* status has shown good concordance with these methods in clinical studies [17,18,20]. No specific assay class developed to detect genomic or protein aberrations will identify all patients with functional PTEN loss and assays are not expected to be fully concordant; therefore the optimal assay for detecting loss of PTEN function is not yet determined.

Ipatasertib is a selective, ATP-competitive, small-molecule inhibitor of all three isoforms of AKT that led to improvements in clinical outcomes in biomarker-selected subgroups of men with mCRPC, with a manageable safety profile [17,18,21–24]. In IPATential150, addition of ipatasertib to abiraterone improved radiographic progression-free survival (rPFS) in patients with PTEN loss on IHC (hazard ratio [HR] 0.77, 95% confidence interval [CI] 0.61–0.98; p = 0.034), but not in the intention-to-treat (ITT) population.

Here we report the final overall survival (OS) analysis of the IPATential150 study and present results for prespecified and exploratory biomarker analyses, including the impact on clinical outcomes of PTEN status assessed via different assays, and genomic alterations in key molecular pathways.

2. Patients and methods

2.1. Study design and patients

IPATential150 (ClinicalTrials.gov NCT03072238) was a randomized, double-blind, phase 3 trial of ipatasertib plus abiraterone versus placebo plus abiraterone as first-line treatment for mCRPC. The study design was previously described [18] (Supplementary material).

The co-primary endpoints were investigator-assessed rPFS in the ITT population and for patients with PTEN loss on IHC. OS was a key secondary endpoint. Additional secondary endpoints included the objective response rate according to Response Evaluation Criteria in Solid Tumors v1.1 and Prostate Cancer Working Group 3 criteria, and the time to prostate-specific antigen (PSA) progression according to Prostate Cancer Working Group 3 criteria.

The study protocol was approved by the relevant institutional review boards or ethics committees at each site. Written informed consent was obtained from all enrolled patients. The clinical trial was performed in accordance with the Declaration of Helsinki.

2.2. Assays

Tumor PTEN status via IHC was centrally assessed using a VENTANA PTEN SP218 assay (Ventana Medical Systems, Oro Valley, AZ, USA). PTEN loss on IHC was defined as no detectable PTEN staining in ≥50% of the tumor area in a specimen [18]. NGS for detection of *PTEN* status and other genomic alterations was performed using a FoundationOne CDx NGS assay (Foundation Medicine, Cambridge, MA, USA) [18]. mRNA levels for transcriptional analysis were assessed in 100 ng of tissue from formalin-fixed, paraffin-embedded (FFPE) samples using TruSeq RNA Access (Illumina, San Diego, CA, USA). Genetic pathway analysis was performed by grouping pathway components as described in Supplementary Table 1.

2.3. Gene set enrichment analysis

Pathway scores were calculated using the single-sample gene set enrichment analysis method [25]. The hallmark gene set collections are from the Molecular Signature database [26]; additional gene sets used for downstream analysis can be found in Supplementary Table 2.

2.4. Statistical analysis

R v4.3.1 (R Foundation for Statistical Computing, Vienna, Austria) was used for statistical analysis. The final OS analyses for patients with PTEN loss on IHC and the ITT population were conducted using SAS v9.4 and were prespecified and α controlled. All other analyses were exploratory. Survival curves were estimated using the Kaplan-Meier method, as previously described [18]. Stratified hazard ratios (sHRs) and 95% confidence intervals (CIs) were estimated using Cox proportional-hazards models. Unstratified HRs are reported for analyses other than ITT and PTEN loss on IHC.

3. Results

The IPATential150 study enrolled 1101 patients, of whom 547 were randomized to ipatasertib + abiraterone and 554 to placebo + abiraterone. Baseline characteristics were generally balanced between the treatment arms (Supplementary Table 3). The representativeness of the study participants is assessed in Supplementary Table 4. Among the enrolled patients, 743 had an NGS assay result, 582 had an RNA sequencing result, and 453 had results for both assays (Supplementary Fig. 1A). OS and rPFS results were consistent between the patient populations defined by assay data availability (Supplementary Fig. 1B–D). The prognostic effect of PTEN loss on IHC was assessed in the placebo arm for rPFS (HR 0.81, 95% CI 0.65–1.02) and OS (HR 0.89, 95% CI 0.72–1.10; Supplementary Fig. 2A). For PTEN loss on NGS, the HR was 0.73 (95% CI, 0.53–1.01) for rPFS and 0.78 (95% CI 0.58–1.06) for OS (Supplementary Fig. 2B). These results are consistent with the known prognostic impact of PTEN loss in mCRPC [11,27].

3.1. Final OS results

At data cutoff (December 31, 2022) for this final OS analysis, median follow-up for the ITT population was 33.9 mo. Among those with PTEN loss on IHC, 158 patients in the ipatasertib + abiraterone arm and 170 in the placebo + abiraterone arm had died, with median OS of 36.8 mo (95% CI 31.4–42.1) and 35.8 mo (95% CI 30.8–39.6), respectively (Fig. 1A). In the ITT population, 318 patients in the ipatasertib + abiraterone arm and 351 in the placebo + abiraterone arm had died, with median OS of 39.4 mo (95% CI 36.5–42.9) and 36.5 mo (95% CI 33.9–39.4), respectively. Addition of ipatasertib to abiraterone did not result in a statistically significant OS benefit for patients with PTEN loss on IHC (sHR 0.94, 95% CI 0.76–1.17; p = 0.57; Fig. 1A). Owing to the statistical testing hierarchy, final OS in the ITT population could not be formally tested (sHR 0.91, 95% CI 0.78–1.06; Fig. 1B).

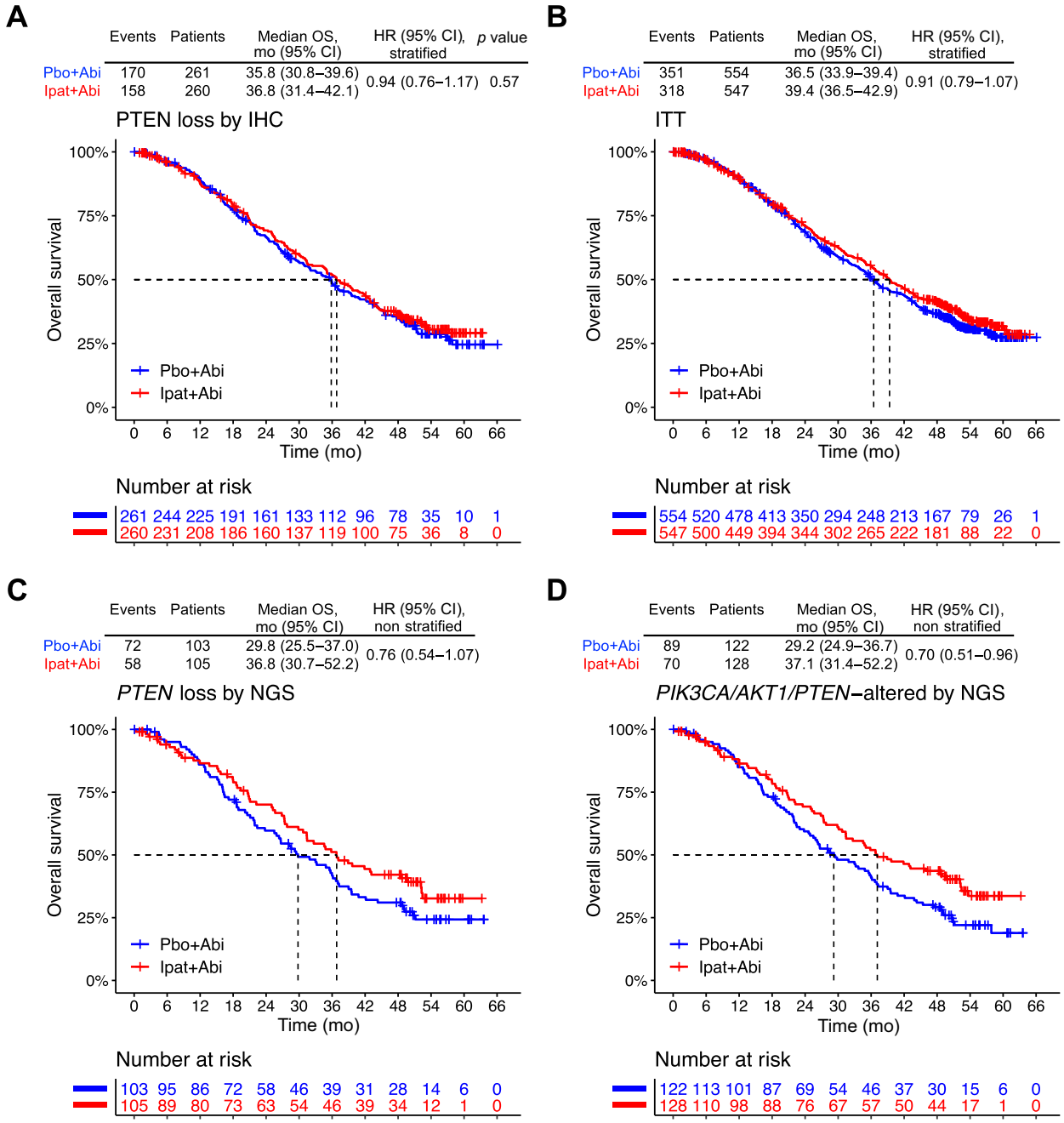


Fig. 1 – OS with ipatasertib + abiraterone versus placebo + abiraterone in IPATential150. Kaplan-Meier curves are shown for (A) the group with PTEN loss on IHC, (B) the ITT population, (C) the group with *PTEN* loss on NGS, and (D) the group with *PIK3CA/AKT1/PTEN* alterations on NGS. Abi = abiraterone; CI = confidence interval; HR = hazard ratio; IHC = immunohistochemistry; Ipat = ipatasertib; ITT = intention to treat; NGS = next-generation sequencing; Pbo = placebo; OS = overall survival.

3.2. *Exploratory PTEN analyses*

PTEN loss on NGS was defined as homozygous deletion, heterozygous deletion, mutation in combination with loss of heterozygosity, or dominant-negative mutation, as previously described [18]. Among patients with an NGS result in this study, 70% (518/743) were evaluable for *PTEN* status.

In the group with *PTEN* loss on NGS, 58 in the ipatasertib + abiraterone arm and 72 in the placebo + abiraterone arm had died (Fig. 1C), with median OS of 36.8 mo (95% CI 30.7–52.2) and 29.8 mo (95% CI 25.5–37.0), respectively (HR 0.76, 95% CI 0.54–1.07). A larger numerical trend for OS benefit was observed when the analysis was expanded to include patients with *PIK3CA/AKT1/PTEN* alterations on NGS (Fig. 1D). In this combined subgroup, 70 patients in the ipatasertib + abiraterone arm and 89 in the placebo + abiraterone arm had died, with median OS of 37.1 mo (95% CI 31.4–52.2) and 29.2 mo (95% CI 24.9–36.7), respectively (HR 0.70, 95% CI, 0.51–0.96). The OS results for these biomarker-selected subgroups are consistent with the clinical rPFS benefit reported for these populations [18].

Among the 1101 patients in the study, the evaluability of *PTEN* status on NGS was consistent across the different subgroups, with different percentages of viable tumor cells with cytoplasmic IHC = 0 (VTCTINT0) scores (Fig. 2A). Concordance between PTEN loss on IHC (defined as a VTCTINT0 score <50) and *PTEN* loss on NGS was 86% (443/518). Among the patients classified as having PTEN loss on IHC, 77% (190/247) had *PTEN* loss on NGS; among the patients with *PTEN* loss on NGS, 91% (190/208) had PTEN loss on IHC (Fig. 2A, bottom). There were trends for higher HR values for rPFS and OS for patients with unknown *PTEN* status on NGS than for patients who had NGS samples evaluable for *PTEN* status (Supplementary Fig. 3A,B), suggesting an imbalance in these populations and highlighting the importance of developing more robust and facile NGS-based methods for *PTEN* assessment.

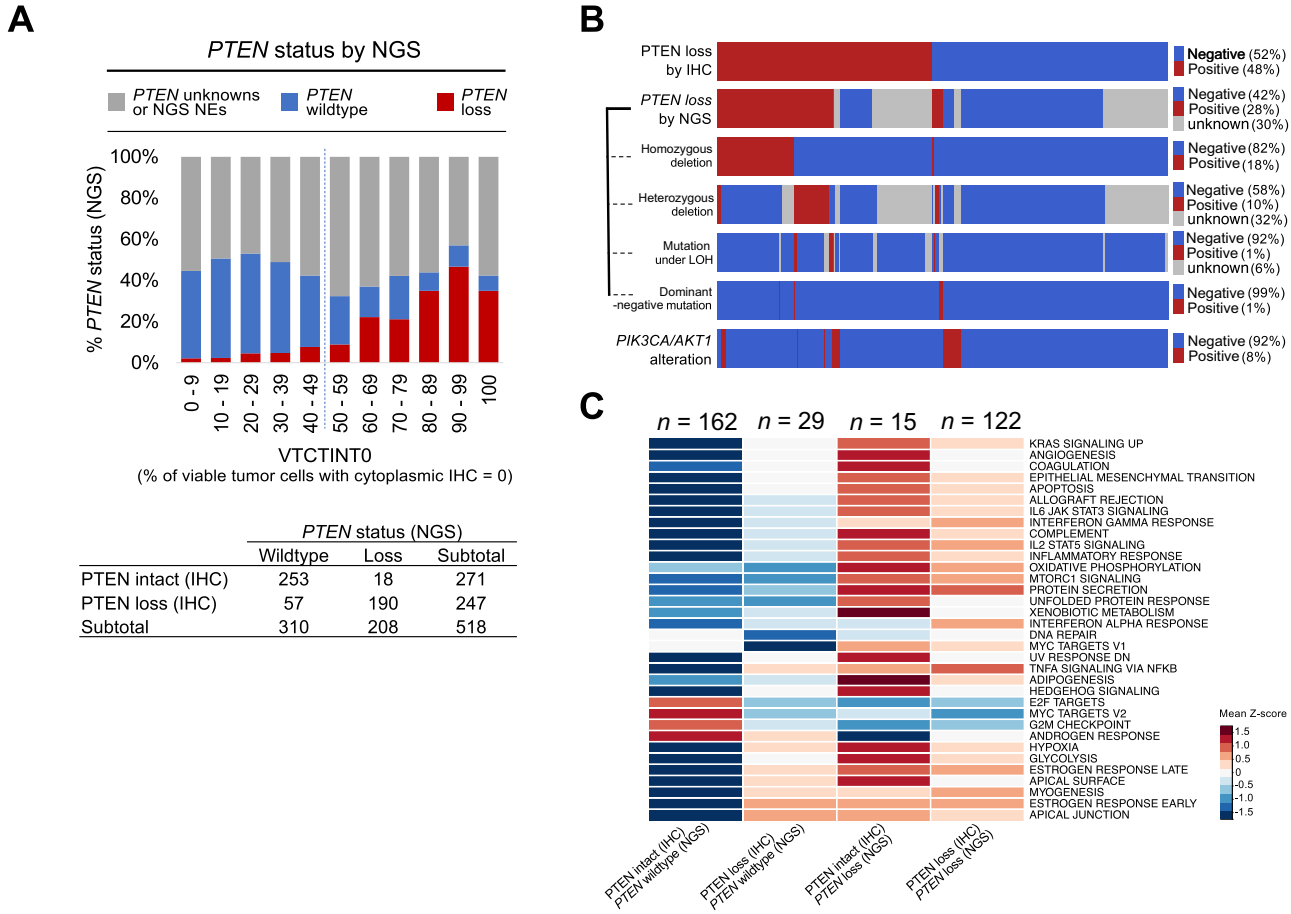
PTEN mRNA levels were lowest in tumor samples with PTEN loss on both IHC and NGS, while higher *PTEN* mRNA levels were observed in samples with PTEN wildtype status or discordant assay results (Supplementary Fig. 3C). A detailed concordance analysis of PTEN status on IHC and NGS (including type of *PTEN* loss) and *PIK3CA*/*AKT1* alteration status is shown in Fig. 2B.

A trend towards better rPFS and OS with addition of ipatasertib was still observed for the 18 patients with intact PTEN according to IHC but *PTEN* loss on NGS (Supplementary Fig. 3D). Among these 18 samples, *PTEN* loss on NGS was because of heterozygous deletion (n = 7), dominant-negative mutation (n = 5), homozygous deletion (n = 4), or mutation classified as loss of heterozygosity (n = 2). However, it is worth noting that different slides were used for IHC and NGS assays without microdissection; therefore, intratumor heterogeneity may have contributed to this discordance in PTEN status between assays. A transcription profile analysis by PTEN assay concordance status is shown in Fig. 2C.

3.3. Genomic alteration analyses

The most frequent single genomic alterations identified in the archival tumor samples were in *TMPRSS2* (31%), *TP53* (30%), and *PTEN* (28%); other mutations identified in ≥5% of samples were in *SPOP*, *RAD21*, *BRCA2*, *PIK3CA*, *APC*, *MYC*, *ATM*, and *CDK12* (Fig. 3A). The frequency of these mutations by ancestry is shown in Supplementary Fig. 4A. Genomic alterations in *TP53* and *TMPRSS2* tended to be enriched in tumors with PTEN loss on either IHC or NGS (both $p \leq 0.001$), while associations of PTEN loss with alterations in *SPOP* ($p \leq 0.01$) and *CDK12* ($p \leq 0.001$) showed strong exclusivity for the assay used (Fig. 3B).

An ad hoc analysis of rPFS and OS by *TMPRSS2*, *TP53*, and *SPOP* status is shown in Fig. 3C–H. In both treatment arms,



	PTEN status (NGS)		
	Wildtype	Loss	Subtotal
PTEN intact (IHC)	253	18	271
PTEN loss (IHC)	57	190	247
Subtotal	310	208	518

Fig. 2 – Analysis of PTEN status via IHC and NGS and transcriptional characterization subgroups. Concordance between (A) PTEN status on IHC and *PTEN* status on NGS and (B) PTEN status by type and *PIK3CA*/*AKT1*/*PTEN* status. (C) mRNA transcription profile in samples with concordant and discordant PTEN status. IHC = immunohistochemistry; LOH = loss of heterozygosity; NE = not evaluable; NGS = next-generation sequencing.

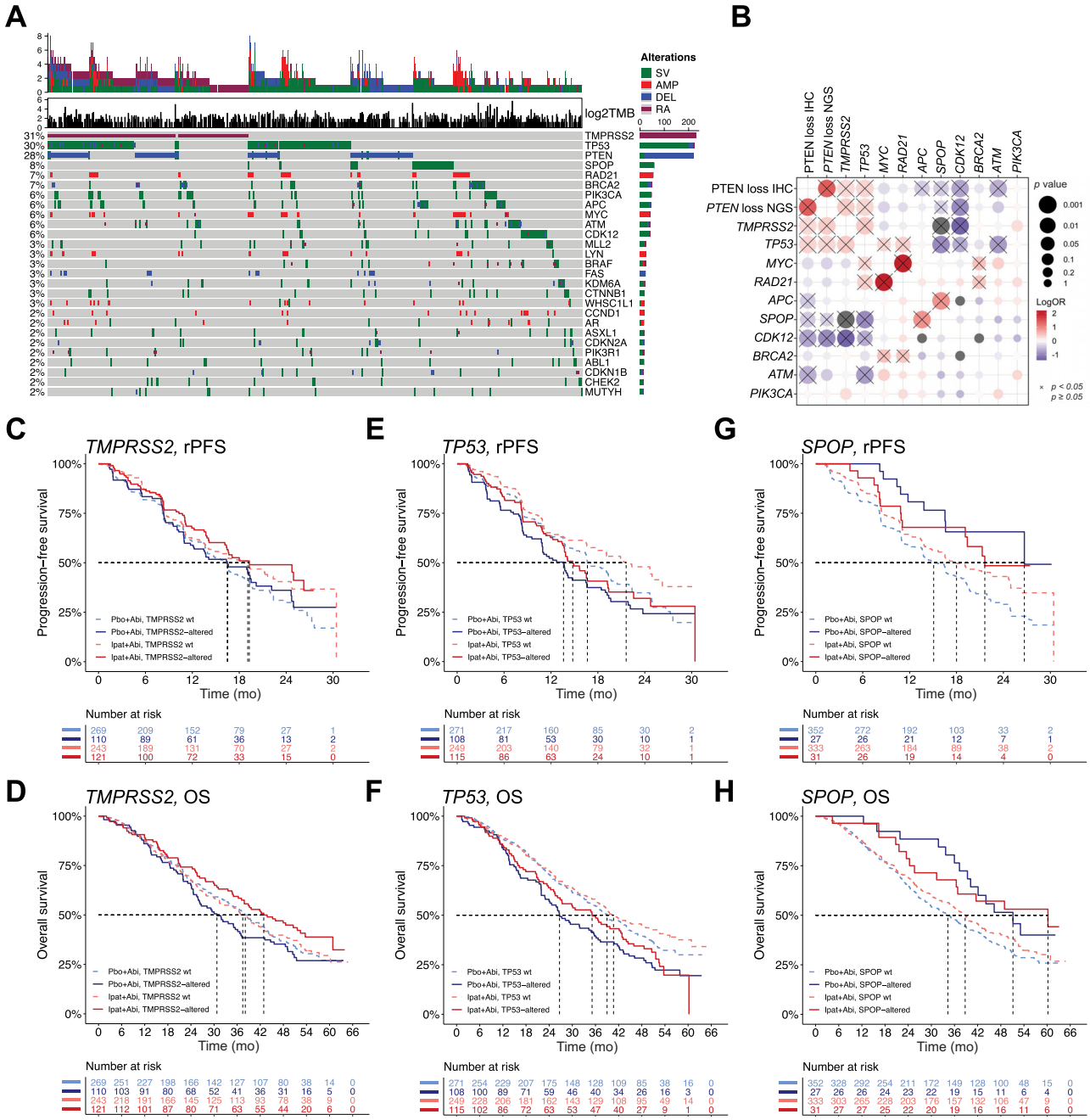


Fig. 3 – Genetic landscape of the patient population and subgroup analysis for the most frequent single genetic alterations. (A) Frequency and co-occurrence of genomic alterations by type and (B) concordance of alterations. Kaplan-Meier estimates of radiographic PFS and OS for samples with or without (C,D) *TMPRSS2*, (E,F) *TP53*, or (G,H) *SPOP* mutations. AMP = amplification; DEL = deletion; IHC = immunohistochemistry; NGS = next-generation sequencing; OS = overall survival; PFS = progression-free survival; RA = rearrangement; rPFS = radiographic progression-free survival; SV = short variant; TMB = tumor mutational burden; wt = wildtype.

patients with *TP53* alterations tended to have shorter rPFS and OS in comparison to those with wildtype *TP53* (Fig. 3E,F), while the inverse relationship was observed between those with and without *SPOP* alterations (Fig. 3G, H). Among patients with *TMPRSS2* alterations, median OS was 42.9 mo (95% CI 36.3–not available) in the ipatasertib + abiraterone arm and 30.8 mo (95% CI, 25.8–37.3) in the placebo + abiraterone arm (Fig. 3C,D and Supplementary Fig. 4B); median rPFS was 19.3 mo (95% CI 16.4–not available) and 16.5 mo (95% CI, 12.1–22.1), respectively. It has been reported that *TMPRSS2*:*ERG* fusion cooperates with PI3K-AKT signaling to drive prostate cancer progression [28,29]. It is therefore possible that tumor cells with *TMPRSS2*:*ERG* fusion might have higher dependence on PI3K-AKT signaling; this hypothesis requires biological and clinical validation.

Further analyses examining different *TP53* mutation types, including dominant-negative mutation (defined as mutants that reduce the transactivation activity of wildtype *TP53* according to the World Health Organization database),

truncation, homozygous deletion, and other missense mutations are shown in Supplementary Fig. 4C.

To evaluate the clinical impact of additional rare genomic alterations, genes were grouped into specific pathways according to their known biological activities (Supplementary Table 1). The pathways most frequently altered were *PIK3CA*/*AKT1*/*PTEN* (34%) and *TP53* (31%). Pathways altered in ≥10% of samples included the homologous recombination repair, cell cycle, and Wnt canonical pathways (Fig. 4A). An rPFS benefit from ipatasertib was not observed for patients with an alteration in the Wnt pathway, in contrast to those without an alteration; a similar trend was

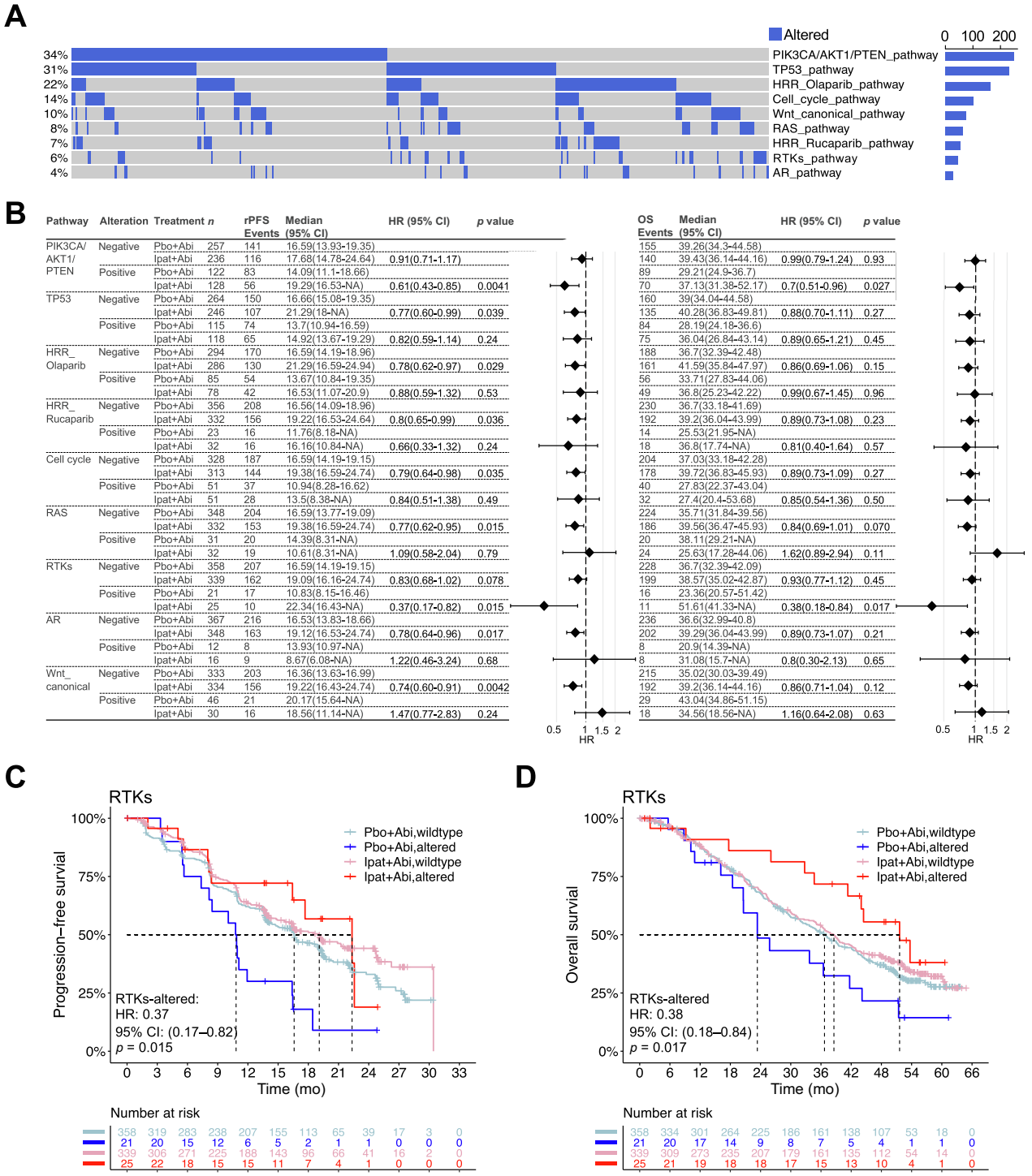


Fig. 4 – Pooling of less frequent mutations into alterations in key genetic pathways for subgroup analysis of the impact of ipatasertib plus abiraterone. (A) Frequency and co-occurrence of pathway alterations. Kaplan-Meier estimates of radiographic PFS and OS by (B) pathway alteration status and (C,D) RTK pathways alteration status. Abi = abiraterone; CI = confidence interval; HR = hazard ratio; IHC = immunohistochemistry; Ipat = ipatasertib; ITT = intention to treat; NA = not available; NGS = next-generation sequencing; Pbo = placebo; PFS = progression-free survival; RTK = receptor tyrosine kinase; OS = overall survival. Bars represent the 95% confidence interval.

observed for OS (Fig. 4B). Of importance, better rPFS and OS with ipatasertib were observed for patients with *PIK3CA*/*AKT1*/*PTEN* alterations and receptor tyrosine kinase (RTK) pathway alterations (Fig. 4B). Among patients with RTK pathways alterations (n = 46), improvements in rPFS (HR 0.37, 95% CI, 0.17–0.82; Fig. 4C) and OS (HR 0.38, 95% CI 0.18–0.84; Fig. 4D) were observed for addition of ipatasertib to abiraterone. No benefit from ipatasertib was observed for patients without RTK pathways alterations (n=697), with HR of 0.83 (95% CI 0.68–1.02) for rPFS and 0.93 (95% CI 0.77–1.12) for OS (Fig. 4B).

3.4. *Prognostic biomarker analysis by gene signature*

Although transcriptome data collected from archival samples may not faithfully capture the disease state of mCRPC, it has been demonstrated that baseline transcriptome analyses can capture relevant disease biology and identify key pathways associated with prognosis in prostate cancer [30]. To evaluate the impact of key biological pathways at baseline on disease progression of mCRPC, we calculated gene signature scores for selected pathways, including luminal and basal subtypes, neuroendocrine prostate cancer, Wnt signaling, DNA damage response, and MYC activity, and took the top and bottom quartiles as high and low groups, respectively. High baseline cell-cycle, DDR, and MYC target activity were the top features associated with poor prognosis (Fig. 5), which was validated using The Cancer Genome Atlas prostate adenocarcinoma data (Supplementary Fig. 5). These findings recapitulate the significance of DDR and MYC biology in disease progression in prostate cancer [31,32].

4. Discussion

This final analysis of the IPATential150 trial reveals that addition of ipatasertib to abiraterone did not lead to a statistically significant improvement in OS for patients with previously untreated mCRPC in both the subgroup with PTEN loss on IHC and the ITT population. As IPATential150 represents one of the largest biomarker studies in men with mCRPC, a detailed biomarker analysis of the study was undertaken to gain a better understanding of prostate cancer biology and to potentially identify a subpopulation of patients who might benefit from treatment with ipatasertib. This analysis further demonstrates the molecular heterogeneity of prostate cancer and identified potentially actionable targets, including PTEN loss, *TMPRSS2* rearrangement, and RTK pathway alterations.

The primary analysis revealed a numerically greater effect of ipatasertib on PFS for patients with genomic *PTEN* loss (HR 0.65, 95% CI 0.45–0.95) or *PIK3CA*/*AK1*/*PTEN* alterations according to NGS (HR 0.63, 95% CI 0.44–0.88) in com-

Gene_ signature	Arm	n	Comparison	rPFS HR (95% CI)	p value	OS HR (95% CI)	p value
AR	Pbo+Abi	149	low vs high	1.17(0.77-1.78)	0.45	1.25(0.84-1.84)	0.27
	Ipat+Abi	143	low vs high	1.13(0.69-1.86)	0.63	1.01(0.66-1.55)	0.95
Basal	Pbo+Abi	144	low vs high	1.16(0.75-1.78)	0.50	1.05(0.70-1.58)	0.81
	Ipat+Abi	148	low vs high	1.16(0.73-1.84)	0.54	1.13(0.74-1.72)	0.56
Cell cycle	Pbo+Abi	153	low vs high	0.48(0.31-0.75)	0.0011	0.54(0.36-0.81)	0.0031
	Ipat+Abi	139	low vs high	0.47(0.29-0.77)	0.0032	0.69(0.44-1.08)	0.10
DDR	Pbo+Abi	145	low vs high	0.49(0.32-0.75)	<0.001	0.59(0.39-0.88)	0.010
	Ipat+Abi	147	low vs high	0.62(0.38-1.02)	0.059	0.69(0.45-1.07)	0.097
Fatty acid and cholesterol biosynthesis	Pbo+Abi	164	low vs high	0.56(0.38-0.85)	0.0062	0.71(0.49-1.04)	0.077
	Ipat+Abi	128	low vs high	0.76(0.47-1.26)	0.29	0.86(0.56-1.32)	0.50
Fibroblast	Pbo+Abi	154	low vs high	1.17(0.78-1.75)	0.46	0.98(0.67-1.44)	0.93
	Ipat+Abi	138	low vs high	1.17(0.72-1.92)	0.52	1.04(0.67-1.62)	0.86
Glucose metabolism	Pbo+Abi	152	low vs high	0.81(0.53-1.22)	0.31	1.07(0.72-1.58)	0.75
	Ipat+Abi	140	low vs high	0.72(0.44-1.18)	0.19	1.06(0.70-1.61)	0.79
IFNg	Pbo+Abi	146	low vs high	0.81(0.53-1.23)	0.32	0.92(0.62-1.37)	0.68
	Ipat+Abi	146	low vs high	0.85(0.53-1.35)	0.48	0.81(0.53-1.24)	0.33
Immune checkpoint	Pbo+Abi	157	low vs high	0.82(0.54-1.23)	0.33	1.03(0.70-1.52)	0.88
	Ipat+Abi	135	low vs high	0.46(0.27-0.78)	0.0042	0.68(0.44-1.05)	0.081
Luminal	Pbo+Abi	142	low vs high	1.03(0.68-1.58)	0.88	1.11(0.74-1.65)	0.62
	Ipat+Abi	150	low vs high	1.3(0.80-2.13)	0.29	1.18(0.78-1.78)	0.44
Macrophage	Pbo+Abi	159	low vs high	0.97(0.66-1.43)	0.88	1.19(0.81-1.74)	0.37
	Ipat+Abi	133	low vs high	0.77(0.46-1.30)	0.33	0.98(0.63-1.52)	0.92
MAPK	Pbo+Abi	163	low vs high	1.05(0.70-1.58)	0.81	0.94(0.65-1.36)	0.73
	Ipat+Abi	129	low vs high	1.18(0.72-1.92)	0.51	1.25(0.80-1.93)	0.33
Neuroendocrine	Pbo+Abi	155	low vs high	1.05(0.70-1.58)	0.82	0.81(0.55-1.20)	0.30
	Ipat+Abi	137	low vs high	0.67(0.41-1.09)	0.11	0.7(0.45-1.07)	0.10
Nucleotide biosynthesis	Pbo+Abi	150	low vs high	0.54(0.36-0.82)	0.0031	0.61(0.41-0.90)	0.013
	Ipat+Abi	142	low vs high	0.51(0.32-0.83)	0.0073	0.71(0.46-1.10)	0.13
RB loss high	Pbo+Abi	151	low vs high	0.63(0.41-0.97)	0.037	0.57(0.38-0.85)	0.0060
	Ipat+Abi	141	low vs high	0.81(0.50-1.29)	0.37	0.77(0.50-1.19)	0.24
RB loss low	Pbo+Abi	148	low vs high	1.15(0.76-1.73)	0.52	0.97(0.64-1.46)	0.87
	Ipat+Abi	144	low vs high	1.17(0.71-1.90)	0.54	1.07(0.71-1.63)	0.74
Teff	Pbo+Abi	151	low vs high	0.8(0.52-1.23)	0.32	1.02(0.68-1.53)	0.93
	Ipat+Abi	141	low vs high	1.21(0.74-1.97)	0.45	1.15(0.75-1.78)	0.52
TGFb	Pbo+Abi	161	low vs high	0.95(0.64-1.40)	0.79	0.94(0.65-1.37)	0.74
	Ipat+Abi	131	low vs high	0.77(0.46-1.27)	0.30	0.77(0.49-1.20)	0.25
Notch signaling	Pbo+Abi	161	low vs high	1.42(0.93-2.18)	0.11	1.2(0.80-1.81)	0.38
	Ipat+Abi	131	low vs high	0.99(0.59-1.65)	0.96	1.12(0.72-1.73)	0.63
Wnt beta catenin signaling	Pbo+Abi	153	low vs high	1.71(1.11-2.62)	0.014	1.33(0.88-2.01)	0.17
	Ipat+Abi	139	low vs high	1.05(0.64-1.72)	0.84	1.21(0.80-1.84)	0.37
Myc targets	Pbo+Abi	150	low vs high	0.48(0.32-0.74)	<0.001	0.49(0.32-0.73)	<0.001
	Ipat+Abi	142	low vs high	0.65(0.40-1.05)	0.082	0.87(0.57-1.34)	0.54

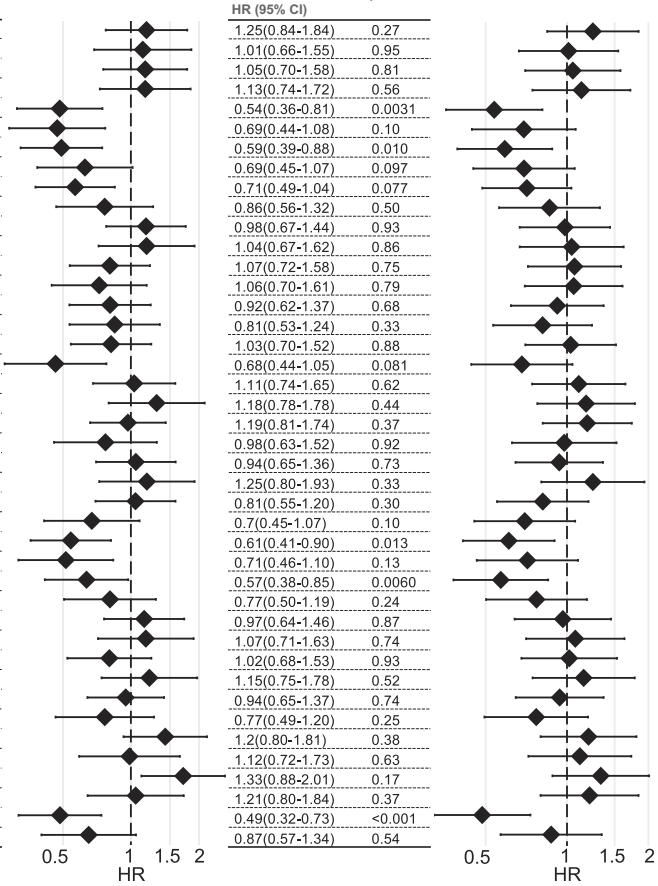


Fig. 5 – Prognostic biomarker analysis. Abi = abiraterone; HR = hazard ratio; Ipat = ipatasertib; OS = overall survival; Pbo = placebo; rPFS = radiographic progression-free survival. Bars represent the 95% confidence interval.

parison to PTEN loss on IHC (HR 0.77, 95% CI 0.61–0.98) [18]. A potential explanation is that despite the level of concordance between the IHC and NGS PTEN assays, there were differences in the patient populations identified as having PTEN loss using these assays, including varying *PTEN* mRNA expression levels and transcription profiles. Overall, these results are consistent with data for patients with hormone receptor–positive breast cancer, for whom addition of the AKT inhibitor capivasertib to fulvestrant improved PFS and OS in the ITT to a greater extent in the subgroup of patients with AKT pathway alterations in their tumor [33].

A potential confounding factor in detecting PTEN loss is its heterogeneity in primary tumors and evolution of this heterogeneity as the disease progresses [10,34]. Tumor heterogeneity can be quantified via IHC assay for multiple parts of a single tumor, whereas NGS assays do not provide insight into tumor *PTEN* heterogeneity as copy numbers are averaged across cells. Assays for PTEN status are currently not part of routine clinical practice for prostate cancer; as PTEN-targeted therapies are developed for prostate cancer, the role of each assay will need to be further assessed. It should also be noted that in the prior supporting phase 2 study of ipatasertib in mCRPC, two different PTEN IHC assays showing 80% concordance were used, but the potential impact of the use of different IHC assays on the results from IPATential150 is unknown [17].

The genes most frequently mutated in our study were *TMPRSS2*, *TP53*, and *PTEN*, and among the genes assessed, these *TMPRSS2* and *TP53* mutations were most likely to co-occur with PTEN loss. By contrast, mutations in *SPOP* were inversely correlated with PTEN loss and with alterations in *TMPRSS2* and *TP53*. The efficacy results observed here are consistent with previously reports in which *TP53* mutations were associated with poor prognosis in prostate cancer [35–37], and *SPOP* mutations were associated with better clinical outcomes [37,38]. Furthermore, mutations in tumor suppressor genes increase with more advanced disease, and the co-occurrence of mutations in *PTEN* and *TMPRSS2* and/or *TP53* is associated with poorer clinical outcomes [39,40]. Mutations in the RTK pathway occurred in 6% of patients in our study, and a better clinical response to ipatasertib was observed for these patients. Although patients with genetic alterations in the RTK pathway represent a small population in mCRPC, RTK functions as a key upstream regulator of PI3K/AKT signaling [41], so these patients might benefit from therapies that inhibit PI3K or AKT.

Strengths of our study include the large population and molecular stratification via multiple methods. However, there are multiple limitations, including the exploratory nature of the NGS analyses, the availability of biomarker data for subsets of patients, and potential heterogeneity confounding the determination of PTEN status. One caveat in this analysis is that most of the biomarker data were collected from archival samples from close to the time of initial disease diagnosis, which may not reflect the mCRPC state because of intratumoral heterogeneity and genomic drift after hormone deprivation therapy. Another limitation is the lack of microdissection because of the limited sample amounts for multiomics analysis. In addition, some samples could not be evaluated via NGS, and among the NGS-evaluable samples, some could not be evaluated for *PTEN* status. Prostate tissue presents challenges for NGS, including limited access to fresh-frozen tissue biopsy samples and technical challenges in sequencing FFPE biopsy samples, especially from bone metastases, as high-complexity NGS assays require more evaluable tissue than IHC assays [42].

5. Conclusions

In conclusion, addition of ipatasertib to abiraterone did not lead to a statistically significant improvement in OS for men with mCRPC with PTEN loss on IHC. Exploratory biomarker analyses, including NGS, identified subpopulations who might benefit from this treatment combination, which could guide the development of drugs targeting the PIK3CA/AKT1/PTEN pathway. These results warrant further investigation in prospective studies.

Author contributions: Christopher Sweeney had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Data sharing statement: For eligible studies, qualified researchers may request access to individual patient-level clinical data through a data request platform. At the time of writing, this request platform is Vivli (https://vivli.org/ourmember/roche/). Up-to-date details on Roche's global policy on the sharing of clinical information and how to request access to related clinical study documents are available at https://go.roche.com/-data_sharing. Anonymized records for individual patients across more than one data source external to Roche cannot, and should not, be linked because of a potential increase in the risk of patient reidentification.

Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eururo.2024.12.015.

References

[1] Smith M, Parker C, Saad F, et al. Addition of radium-223 to abiraterone acetate and prednisone or prednisolone in patients with castration-resistant prostate cancer and bone metastases (ERA 223): a randomised, double-blind, placebo-controlled, phase 3 trial. Lancet Oncol 2019;20:408–19.
[2] Beer TM, Armstrong AJ, Rathkopf DE, et al. Enzalutamide in metastatic prostate cancer before chemotherapy. N Engl J Med 2014;371:424–33.
[3] Moreira DM, Howard LE, Sourbeer KN, et al. Predicting time from metastasis to overall survival in castration-resistant prostate cancer: results from SEARCH. Clin Genitourin Cancer 2017;15, 60–6.e2.
[4] Antonarakis ES, Armstrong AJ, Dehm SM, Luo J. Androgen receptor variant-driven prostate cancer: clinical implications and therapeutic targeting. Prostate Cancer Prostat Dis 2016;19:231–41.
[5] Labrecque MP, Coleman IM, Brown LG, et al. Molecular profiling stratifies diverse phenotypes of treatment-refractory metastatic castration-resistant prostate cancer. J Clin Invest 2019;129:4492–505.
[6] Brady L, Kriner M, Coleman I, et al. Inter- and intra-tumor heterogeneity of metastatic prostate cancer determined by digital spatial gene expression profiling. Nat Commun 2021;12:1426.
[7] Massard C, Oulhen M, Le Moulec S, et al. Phenotypic and genetic heterogeneity of tumor tissue and circulating tumor cells in patients with metastatic castration-resistant prostate cancer: a report from the PETRUS prospective study. Oncotarget 2016;7:55069–82.
[8] Tang F, Xu D, Wang S, et al. Chromatin profiles classify castration-resistant prostate cancers suggesting therapeutic targets. Science 2022;376:eabe1505.
[9] Gasmi A, Roubaud G, Dariane C, et al. Overview of the development and use of Akt inhibitors in prostate cancer. J Clin Med. 2021;11.
[10] Jamaspishvili T, Berman DM, Ross AE, et al. Clinical implications of PTEN loss in prostate cancer. Nat Rev Urol 2018;15:222–34.
[11] Ferraldeschi R, Nava Rodrigues D, Riisnaes R, et al. PTEN protein loss and clinical outcome from castration-resistant prostate cancer treated with abiraterone acetate. Eur Urol 2015;67:795–802.
[12] Gupta S, To TM, Graf R, Kadel 3rd EE, Reilly N, Albarmawi H. Real-world overall survival and treatment patterns by PTEN status in metastatic castration-resistant prostate cancer. JCO Precis Oncol 2024;8:e2300562.
[13] Reid AH, Attard G, Brewer DJ, et al. Novel, gross chromosomal alterations involving PTEN cooperate with allelic loss in prostate cancer. Mod Pathol 2012;25:902–10.
[14] Correia NC, Girio A, Antunes I, Martins LR, Barata JT. The multiple layers of non-genetic regulation of PTEN tumour suppressor activity. Eur J Cancer 2014;50:216–25.
[15] Verhagen PC, van Duijn PW, Hermans KG, et al. The PTEN gene in locally progressive prostate cancer is preferentially inactivated by bi-allelic gene deletion. J Pathol 2006;208:699–707.
[16] Vidotto T, Melo CM, Castelli E, Koti M, Dos Reis RB, Squire JA. Emerging role of PTEN loss in evasion of the immune response to tumours. Br J Cancer 2020;122:1732–43.
[17] de Bono JS, De Giorgi U, Rodrigues DN, et al. Randomized phase II study evaluating Akt blockade with ipatasertib, in combination with abiraterone, in patients with metastatic prostate cancer with and without PTEN loss. Clin Cancer Res 2019;25:928–36.
[18] Sweeney C, Bracarda S, Sternberg CN, et al. Ipatasertib plus abiraterone and prednisolone in metastatic castration-resistant prostate cancer (IPATential150): a multicentre, randomised, double-blind, phase 3 trial. Lancet 2021;398:131–42.
[19] Lotan TL, Wei W, Ludkovski O, et al. Analytic validation of a clinical-grade PTEN immunohistochemistry assay in prostate cancer by comparison with PTEN FISH. Mod Pathol 2016;29:904–14.

[20] de Bono J, Sweeney C, Bracarda S, et al. PI3K/AKT pathway biomarkers analysis from the phase III IPATential150 trial of ipatasertib plus abiraterone in metastatic castration resistant prostate cancer. J Clin Oncol 2021;396(Suppl):13.
[21] Lin J, Sampath D, Nannini MA, et al. Targeting activated Akt with GDC-0068, a novel selective Akt inhibitor that is efficacious in multiple tumor models. Clin Cancer Res 2013;19:1760–72.
[22] Blake JF, Xu R, Bencsik JR, et al. Discovery and preclinical pharmacology of a selective ATP-competitive Akt inhibitor (GDC-0068) for the treatment of human tumors. J Med Chem 2012;55:8110–27.
[23] Matsubara N, de Bono J, Sweeney C, et al. Safety profile of ipatasertib plus abiraterone vs placebo plus abiraterone in metastatic castration-resistant prostate cancer. Clin Genitourin Cancer 2023;21, 230–7.e1.
[24] Kotani N, Wilkins JJ, Wade JR, et al. Characterization of exposure-response relationships of ipatasertib in patients with metastatic castration-resistant prostate cancer in the IPATential150 study. Cancer Chemother Pharmacol 2022;90:511–21.
[25] Barbie DA, Tamayo P, Boehm JS, et al. Systematic RNA interference reveals that oncogenic KRAS-driven cancers require TBK1. Nature 2009;462:108–12.
[26] Subramanian A, Tamayo P, Mootha VK, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. Proc Natl Acad Sci U S A 2005;102:15545–50.
[27] Lotan TL, Heumann A, Rico SD, et al. PTEN loss detection in prostate cancer: comparison of PTEN immunohistochemistry and PTEN FISH in a large retrospective prostatectomy cohort. Oncotarget 2017;8:65566–76.
[28] Carver BS, Tran J, Gopalan A, et al. Aberrant ERG expression cooperates with loss of PTEN to promote cancer progression in the prostate. Nat Genet 2009;41:619–24.
[29] King JC, Xu J, Wongvipat J, et al. Cooperativity of TMPRSS2-ERG with PI3-kinase pathway activation in prostate oncogenesis. Nat Genet 2009;41:524–6.
[30] Bolis M, Bossi D, Vallerga A, et al. Dynamic prostate cancer transcriptome analysis delineates the trajectory to disease progression. Nat Commun 2021;12:7033.
[31] Schiewer MJ, Knudsen KE. DNA damage response in prostate cancer. Cold Spring Harb Perspect Med 2019;9:a030486.
[32] Koh CM, Bieberich CJ, Dang CV, Nelson WG, Yegnasubramanian S, De Marzo AM. MYC and prostate cancer. Genes. Cancer 2010;1:617–28.
[33] Turner NC, Oliveira M, Howell SJ, et al. Capivasertib in hormone receptor-positive advanced breast cancer. N Engl J Med 2023;388:2058–70.
[34] Krohn A, Freudenthaler F, Harasimowicz S, et al. Heterogeneity and chronology of PTEN deletion and ERG fusion in prostate cancer. Mod Pathol 2014;27:1612–20.
[35] Sun J, Zhang K, Cai Z, et al. Identification of critical pathways and hub genes in TP53 mutation prostate cancer by bioinformatics analysis. Biomark Med 2019;13:831–40.
[36] Ecke TH, Schlechte HH, Schiemenz K, et al. TP53 gene mutations in prostate cancer progression. Anticancer Res 2010;30:1579–86.
[37] Annala M, Vandekerkhove G, Khalaf D, et al. Circulating tumor DNA genomics correlate with resistance to abiraterone and enzalutamide in prostate cancer. Cancer Discov 2018;8:444–57.
[38] Zhou J, Lai Y, Peng S, et al. Comprehensive analysis of TP53 and SPOP mutations and their impact on survival in metastatic prostate cancer. Front Oncol 2022;12:957404.
[39] Hamid AA, Gray KP, Shaw G, et al. Compound genomic alterations of TP53, PTEN, and RB1 tumor suppressors in localized and metastatic prostate cancer. Eur Urol 2019;76:89–97.
[40] Brady L, Carlsson J, Baird AM, et al. Correlation of integrated ERG/PTEN assessment with biochemical recurrence in prostate cancer. Cancer Treat Res Commun 2021;29:100451.
[41] He Y, Sun MM, Zhang GG, et al. Targeting PI3K/Akt signal transduction for cancer therapy. Signal Transduct Target Ther 2021;6:425.
[42] Hovelson DH, Tomlins SA. The role of next-generation sequencing in castration-resistant prostate cancer treatment. Cancer J 2016;22:357–61.